

AKUNE, Japan

WMO: 478230

Lat: 32.0278N

Lon: 130.2000E

Elev: 42

StdP: 100.82

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.5	2.6	-5.8	2.3	3.9	-4.6	2.6	4.7	9.8	8.6	8.8	8.3	4.2	70	0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.1	32.1	26.0	31.1	26.1	30.3	25.9	27.2	30.1	26.7	29.6	26.4	29.2	3.4	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.5	22.1	29.0	26.0	21.4	28.7	25.5	20.8	28.4	86.4	30.2	84.3	29.8	82.5	29.4	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.1	6.2	-0.4	34.4	1.1	0.7	-1.2	34.9	-1.9	35.3	-2.5	35.7	-3.3	36.3		
			-2.4	27.9	1.1	0.6	-3.1	28.3	-3.7	28.6	-4.3	28.9	-5.1	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	17.7	7.8	8.8	11.5	15.7	19.7	22.9	26.9	27.8	25.1	20.5	15.2	10.1	(6)
(7)		DBStd	7.30	2.84	3.26	3.02	2.74	2.01	2.04	1.64	1.39	2.16	2.55	3.14	3.16	(7)
(8)		HDD10.0	194	79	57	18	1	0	0	0	0	0	1	38	(8)	
(9)		HDD18.3	1262	326	267	211	85	9	0	0	0	8	102	255	(9)	
(10)		CDD10.0	3010	12	24	66	173	299	387	523	551	452	325	157	42	(10)
(11)		CDD18.3	1036	0	0	1	7	50	137	264	293	202	75	8	0	(11)
(12)		CDH23.3	8206	0	0	1	9	104	601	2566	3158	1502	258	7	0	(12)
(13)		CDH26.7	2271	0	0	0	0	6	72	716	1095	357	26	0	0	(13)
(14)	Wind	WSAvg	3.2	3.8	3.7	3.7	3.2	2.7	2.6	2.7	2.6	2.9	3.2	3.3	3.7	(14)
(15)	Precipitation	PrecAvg	2139	79	98	134	160	186	453	319	214	201	97	104	93	(15)
(16)		PrecMax	3435	157	204	330	395	369	1018	926	670	449	284	215	170	(16)
(17)		PrecMin	1342	17	40	63	81	58	88	51	28	30	9	31	7	(17)
(18)		PrecStd	464	30	46	54	73	92	199	206	135	132	70	50	42	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	19.8	21.5	24.6	27.5	30.2	33.0	34.0	32.1	28.9	24.2	19.8	(19)
(20)			MCWB	15.5	16.8	16.3	18.7	21.0	25.5	25.9	25.9	25.1	23.7	19.7	16.2	(20)
(21)		2%	DB	15.9	17.9	19.6	22.8	25.6	28.6	31.5	32.6	30.6	27.2	22.7	18.2	(21)
(22)			MCWB	13.2	15.4	15.8	18.1	20.4	24.7	26.2	26.0	25.3	22.4	18.6	15.1	(22)
(23)		5%	DB	14.2	16.3	18.2	21.6	24.4	27.2	30.6	31.6	29.7	25.9	21.4	16.6	(23)
(24)			MCWB	11.2	13.8	15.0	17.7	19.9	23.9	26.2	26.2	24.8	21.1	17.8	13.3	(24)
(25)		10%	DB	12.6	14.5	16.9	20.4	23.4	26.2	29.9	30.8	28.7	24.7	20.1	15.1	(25)
(26)			MCWB	9.4	11.4	13.6	16.6	19.4	23.4	25.9	26.1	24.1	20.3	16.8	12.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	17.9	18.4	21.3	23.5	26.4	27.9	28.1	26.8	25.1	21.5	17.7	(27)
(28)			MCDWB	17.6	19.0	19.6	22.5	25.3	28.9	30.3	30.8	29.6	27.5	23.0	18.8	(28)
(29)		2%	WB	13.8	15.9	16.8	19.8	22.2	25.6	27.2	27.4	26.2	23.6	19.8	15.6	(29)
(30)			MCDWB	15.5	17.5	18.6	21.4	24.0	27.7	29.7	30.4	29.2	26.2	21.4	17.6	(30)
(31)		5%	WB	11.6	14.1	15.5	18.5	21.2	25.0	26.7	26.9	25.6	22.4	18.6	13.8	(31)
(32)			MCDWB	13.7	16.1	17.8	20.6	23.1	26.6	29.3	30.1	28.6	25.0	20.7	16.0	(32)
(33)		10%	WB	9.8	11.9	14.1	17.4	20.2	24.3	26.3	26.5	25.0	21.2	17.3	12.4	(33)
(34)			MCDWB	12.1	14.1	16.6	19.7	22.5	25.8	28.9	29.7	28.0	23.9	19.7	14.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.1	6.7	7.2	7.2	6.9	5.4	5.3	6.1	6.4	6.8	6.6	6.3	(35)
(36)			MCDBR	8.4	8.5	8.6	8.4	8.0	6.3	6.3	7.1	7.0	7.7	7.4	7.7	(36)
(37)			MCWBR	6.9	7.4	6.4	5.6	4.9	3.5	3.1	3.3	3.5	4.2	5.0	5.9	(37)
(38)		5% WB	MCDBR	8.0	7.7	7.6	6.6	6.0	4.6	5.5	6.1	6.2	6.1	6.0	7.1	(38)
(39)			MCWBR	7.4	7.5	6.7	5.5	4.7	3.1	3.0	3.1	3.4	4.0	5.0	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.433	0.468	0.524	0.543	0.552	0.554	0.537	0.521	0.502	0.463	0.430	0.423	(40)	
(41)		taud	2.079	2.017	1.882	1.891	1.936	2.020	2.106	2.149	2.155	2.174	2.183	2.112	(41)	
(42)		Ebn at Noon	758	773	760	766	762	757	768	775	772	772	760	743	(42)	
(43)		Edh at Noon	133	156	192	198	191	175	160	151	144	132	120	123	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.26	2.97	4.03	4.79	5.04	4.17	5.27	5.40	4.46	3.71	2.69	2.13	(44)	
(45)		RadStd	0.23	0.28	0.27	0.44	0.56	0.68	0.65	0.42	0.47	0.39	0.26	0.20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page